

Irrational areas of bounded remainder windows

A literal-dimension counterexample to TLMC 00000007701

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Abstract

We construct a rank-three cut-and-project lattice with physical space $\mathbb{R}$ and internal space $\mathbb{R}^2$, preserving the conjecture's explicit window dimension. Its physical projection is injective and its internal projection is dense. The rectangles $W_q = [0, \{-q\sqrt{2}\}) \times [0, 1)$ are bounded remainder windows: the discrepancy on every physical interval is at most $q + 2$. Their actual Lebesgue areas are irrational and form a dense subset of $(0, 1)$. Thus the claimed half-integer and discrete volume spectrum is false. Lean verifies the lattice parametrization, model-set definition, exact physical counting on every real interval, bounded remainder window membership, the universal half-integer volume-clause negation, and area density within that physical class. The only human bridge needed for the literal disproof is lattice admissibility: full rank and covolume, physical injectivity, and internal density.

1 Statement, dimensions, and counting convention

The conjecture is reproduced verbatim:

Definition: For an irrational rotation T_α on the circle, a set B is a bounded remainder set (BRS) if the Birkhoff sums of its indicator minus $n|B|$ are uniformly bounded; higher-dimensional BRS are built by the cut-and-project construction with a window W in $\mathbb{R}^m$ and lattice Γ . Conjecture: W is a d -dimensional BRS if and only if W is a polytope defined by integer-coefficient half-spaces in $\mathbb{R}^m$; and for $d=1$, $m=2$ the volume spectrum of W is a discrete set with all values in $(1/2)\mathbb{Z}$. (polytopal characterization of bounded remainder sets)

We take $d = 1$ to be the physical dimension and $m = 2$ to be the internal dimension: $W \subset \mathbb{R}^2$ and $\Gamma \subset \mathbb{R} \times \mathbb{R}^2$. Our primary construction uses exactly these dimensions. A secondary interpretation with total lattice dimension two is discussed separately in Section 7; we make no assertion that it is the usual meaning of the conjecture's m .

A cut-and-project scheme here consists of a full-rank lattice Γ in $\mathbb{R} \times \mathbb{R}^2$, the coordinate projections π_{phys} and π_{int} , injectivity of $\pi_{\text{phys}}|_\Gamma$, and density of $\pi_{\text{int}}(\Gamma)$. For a bounded measurable window W , define

$$\Lambda(W) = \{\pi_{\text{phys}}(\gamma) : \gamma \in \Gamma, \pi_{\text{int}}(\gamma) \in W\}.$$

Write $c = \text{covol}(\Gamma)$ and $\text{dens}(\Gamma) = 1/c$. We call W a bounded remainder window when $\Lambda(W)$ is locally finite and a constant C satisfies

$$|\#(\Lambda(W) \cap [x, y)) - \text{dens}(\Gamma) \text{vol}_2(W)(y - x)| \leq C \quad (x, y \in \mathbb{R}, x \leq y). \quad (1)$$

Counting is of distinct physical points, with both positive and negative lattice indices allowed. Intervals are bounded and half-open; the empty interval is included. This is the physical dimension-one bounded counting-discrepancy formulation of the bounded-distance-to-a-lattice notion used in

cut-and-project theory. Grepstad–Lev [2] discuss the relation between rotation BRS and cut-and-project bounded distance; Haynes–Kelly–Koivusalo [3] construct BRS and corresponding cut-and-project sets bounded distance to lattices. We prove the needed relation directly, rather than invoke an equivalence without checking its hypotheses. Grepstad–Lev’s rotation definition is uniform in orbit length for almost every starting point; our elementary examples satisfy the stronger bound for every starting point.

The windows below have boundary of planar measure zero. Formula (1) also proves, without needing a general model-set density theorem, that $\Lambda(W)$ has density $\text{dens}(\Gamma) \text{vol}_2(W)$. The area in the conjecture is $\text{vol}_2(W)$ itself; it is not the physical point density.

2 A rank-three admissible lattice

Set $\alpha = \sqrt{2}$, $\beta = \sqrt{3}$, and define

$$A = \begin{pmatrix} 1 & \alpha & \beta \\ -\alpha & 1 & 0 \\ -\beta & 0 & 1 \end{pmatrix}, \quad \Gamma = AZ^3.$$

Thus the three generators are $(1, -\alpha, -\beta)$, $(\alpha, 1, 0)$, and $(\beta, 0, 1)$, and

$$\gamma(k, m, n) = (k + \alpha m + \beta n, m - k\alpha, n - k\beta). \quad (2)$$

Lemma 1. *This is a cut-and-project scheme with physical space $\mathbb{R}$ and internal space $\mathbb{R}^2$, of lattice covolume $c = 6$ and lattice density $1/6$.*

Proof. The determinant of A is $1 + \alpha^2 + \beta^2 = 6$. Therefore A is invertible, AZ^3 is discrete and full rank, its three generators are a lattice basis, and its fundamental parallelepiped has volume 6.

The numbers $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$. Here is an elementary verification. If $k + m\sqrt{2} + n\sqrt{3} = 0$ with integer coefficients and $n = 0$, irrationality of $\sqrt{2}$ gives $k = m = 0$. If $n \neq 0$, squaring gives

$$3n^2 = k^2 + 2m^2 + 2km\sqrt{2}.$$

If $km \neq 0$, this makes $\sqrt{2}$ rational, a contradiction. If $m = 0$, then $3n^2 = k^2$, which is impossible for $n \neq 0$ by parity of the prime-3 valuation. If $k = 0$, then $3n^2 = 2m^2$, also impossible by parity of the prime-3 valuation (and $m = 0$ would force $n = 0$). Consequently all coefficients vanish. The difference of two lattice parameters with equal physical coordinates thus vanishes, proving physical injectivity.

The internal image is $\mathbb{Z}^2 + \mathbb{Z}(-\alpha, -\beta)$. By Kronecker’s density theorem, the orbit $k(\alpha, \beta)$ modulo $\mathbb{Z}^2$ is dense precisely when $1, \alpha, \beta$ are rationally independent, as just verified. Adding arbitrary integer coordinates lifts this density to $\mathbb{R}^2$. In particular the torus rotation is minimal; the full second-coordinate width of the windows is not a consequence of a zero or rational second rotation parameter. $\square$

3 Circle discrepancy used in the counting bridge

For any real r and nonnegative integer q , put

$$a_q = \{qr\}, \quad I_q = [0, a_q), \quad F_q(x) = \sum_{j=0}^{q-1} \{x - (j+1)r\}.$$

Lemma 2. *For every $x \in \mathbb{R}$ and $N \geq 0$,*

$$\sum_{j=0}^{N-1} \mathbf{1}_{I_q}(\{x + jr\}) - Na_q = F_q(x) - F_q(x + Nr), \quad \left| \sum_{j=0}^{N-1} \mathbf{1}_{I_q}(\{x + jr\}) - Na_q \right| \leq q. \quad (3)$$

Proof. Fractional-part subtraction satisfies

$$\{x - t\} = \{x\} - \{t\} + \mathbf{1}_{\{\{x\} < \{t\}\}}.$$

Indeed the difference of the two fractional parts lies in $(-1, 1)$, and adding one is necessary exactly when it is negative. The strict comparison includes equality correctly for a half-open interval. Cancellation in the finite sum gives

$$F_q(x) - F_q(x + r) = \{x - qr\} - \{x\} = \mathbf{1}_{I_q}(\{x\}) - a_q.$$

Telescoping along N successive iterates proves the identity in (3). Every fractional part belongs to $[0, 1)$, so $0 \leq F_q \leq q$ and the absolute difference of two such values is at most q . $\square$

This is the classical Hecke–Ostrowski interval construction. Kesten’s interval theorem [1] characterizes interval BRS lengths by membership in $\mathbb{Z} + r\mathbb{Z}$ for irrational r . We use only the elementary sufficient direction proved above.

4 Exact physical counting and arbitrary intervals

From now on $r = -\alpha = -\sqrt{2}$ and

$$W_q = [0, \{-q\alpha\}) \times [0, 1), \quad q \geq 1, \quad a_q = \{-q\alpha\}.$$

An internal coordinate $(m - k\alpha, n - k\beta)$ lies in $[0, 1)^2$ if and only if

$$m = -\lfloor -k\alpha \rfloor, \quad n = -\lfloor -k\beta \rfloor. \quad (4)$$

This holds for every integer k , including negative ones: it is just uniqueness of the representative modulo $\mathbb{Z}$ in $[0, 1)$. For these unique parameters write

$$u_k = \{-k\alpha\}, \quad v_k = \{-k\beta\}, \quad t_k = ck + \alpha u_k + \beta v_k. \quad (5)$$

Substitution in (2) proves (5), with internal coordinate (u_k, v_k) . The model set is consequently exactly

$$\Lambda(W_q) = \{t_k : k \in \mathbb{Z}, u_k < a_q\}. \quad (6)$$

Since $0 \leq \alpha u_k + \beta v_k < \alpha + \beta < 6 = c$, we have

$$t_k \in [ck, c(k+1)). \quad (7)$$

There is at most one model point in each of these disjoint cells, and only finitely many cells meet a bounded interval. Thus local finiteness and distinctness of counted points hold directly.

Lemma 3. *For every $K \in \mathbb{Z}$ and $N \in \mathbb{N}$,*

$$\#(\Lambda(W_q) \cap [cK, c(K+N))) = \sum_{j=0}^{N-1} \mathbf{1}_{I_q}(\{-K\alpha - j\alpha\}), \quad (8)$$

and the difference between either side and Na_q is at most q .

Proof. By (7), a selected point lies in that physical interval if and only if $K \leq k < K + N$. Reindex by $k = K + j$. The selection condition in (6) is exactly the indicator in (8). Apply (3) with $x = -K\alpha$ and $r = -\alpha$. Also $Na_q = (a_q/c)cN$, with the correct lattice-density normalization. The counting reduction uses the actual physical positions (5), not only an abstract torus predicate. $\square$

Theorem 4. *For the admissible scheme of Lemma 1, every W_q is a bounded remainder window in the sense of (1), with bound $q + 2$ and physical density $a_q/6$.*

Proof. Fix $x \leq y$ and put $\ell = (y - x)/6$. For the outer block set $K = \lfloor x/6 \rfloor$, $J = \lceil y/6 \rceil$, and $N = J - K \geq 0$. Then

$$[x, y] \subseteq [6K, 6J], \quad N - \ell \leq 2.$$

The outer model-set intersection is finite by Lemma 3, so the intersection with $[x, y]$ is finite too. Monotonicity of finite-set cardinality and that lemma imply, writing $A = \#(\Lambda(W_q) \cap [x, y])$,

$$A - a_q \ell \leq q + a_q(N - \ell) \leq q + 2.$$

For the inner block set $L = \lceil x/6 \rceil$, $H = \lfloor y/6 \rfloor$. If $L \leq H$, put $M = H - L$. Then

$$[6L, 6H] \subseteq [x, y], \quad \ell - M \leq 2,$$

and the same count monotonicity gives $A - a_q \ell \geq -q - a_q(\ell - M) \geq -q - 2$. If $H < L$, take the empty aligned block $[6L, 6L]$; the interval has $\ell < 2$, and $A \geq 0$ gives the same lower bound. Thus in all cases

$$\left| \#(\Lambda(W_q) \cap [x, y]) - \frac{a_q}{6}(y - x) \right| \leq q + 2.$$

The nonempty inner and outer blocks differ by at most two lattice steps; when the inner block is empty, the outer block has at most two steps. All constants are independent of x, y . This entire sandwich proof, including finiteness and the case without a complete cell, is formalized in `literal_window_discrepancy`. Measurability of the rectangle then gives `literal_window_isPhysicalBRS`. Dividing by $y - x$ as the length tends to infinity yields density $a_q/6$. $\square$

Our formal witnesses use half-open windows. Replacing W_q by its closed rectangle leaves the area unchanged and alters the entire model set by finitely many points. Indeed each boundary coordinate equation $m - k\alpha \in \{0, a_q\}$ or $n - k\beta \in \{0, 1\}$ fixes at most one integer k , by irrationality, and the remaining bounded coordinate allows only finitely many integers. Thus a closed-window convention also gives bounded discrepancy. The same observation handles endpoint choices for physical intervals, with an additional count error of at most two.

5 Irrational area and the nondiscrete spectrum

Theorem 5. *In the physical- $\mathbb{R}$, internal- $\mathbb{R}^2$ scheme (2),*

$$\neg(\forall W \text{ a bounded remainder window, } \text{vol}_2(W) \in \tfrac{1}{2}\mathbb{Z}).$$

Furthermore its BRS area spectrum is not discrete. In particular the conjecture's conjunction is false under its literal dimensions.

Proof. Product Lebesgue measure gives $\text{vol}_2(W_q) = a_q = \{-q\sqrt{2}\}$. For $q \geq 1$, the number $-q\sqrt{2}$ is irrational; subtracting its integer floor preserves irrationality. Hence a_q is irrational, lies strictly between zero and one, and cannot equal an integer divided by two. Theorem 4 supplies the required membership in the cut-and-project BRS class, already for $q = 1$. The Lean headline theorem `literal_volume_clause_false` states precisely this universal negation over `IsPhysicalBRS`.

The positive multiples of an irrational circle element are dense in $\mathbb{R}/\mathbb{Z}$. Thus for every $0 \leq a < b \leq 1$, some $q \geq 1$ has $a < \{-q\sqrt{2}\} < b$. The source proves this as an ordinary real open-interval statement, using Mathlib's circle-orbit density and unique representatives in $[0, 1)$. The declaration `literal_physical_brs_areas_dense` includes physical BRS membership of every witness in this density statement. The areas of this family are therefore dense in $(0, 1)$, with closure $[0, 1]$. If a spectrum point $v \in (0, 1)$ were isolated, a sufficiently small interval $(v, v + \varepsilon)$ would contain no spectrum point, contradicting this density. This proves nondiscreteness of the full physical BRS area spectrum. We do not assert that its full closure is $[0, 1]$, since other windows can have larger area. $\square$

No polytopal classification is needed to negate the asserted conjunction. As an additional human observation, a bounded planar polytope defined by half-spaces with both integer normals and integer offsets has rational vertices (solve the active linear equations), hence rational area by triangulation. The irrational area of W_q excludes that reading of the geometric characterization, even up to null sets. If arbitrary real offsets are permitted, this particular geometric argument does not apply; the independent area-spectrum contradiction still holds.

6 Precise formalization boundary

The single Lean file imports Mathlib, uses actual Lebesgue volume, and contains the following definitions:

```
Index3 = Int * (Int * Int)
Ambient3 = Real * (Real * Real)
gamma alpha beta (k,m,n) =
  (k + alpha*m + beta*n, (m-k*alpha, n-k*beta))
lattice alpha beta = Set.range (gamma alpha beta)
phys p = p.1
internal p = p.2
modelSet alpha beta W =
  phys '' {p in lattice alpha beta | internal p in W}
covol alpha beta = 1 + alpha^2 + beta^2
physicalCount alpha beta W x y =
  (modelSet alpha beta W intersect Set.Ico x y).ncard
```

The actual identifiers for the first two types use subscript 3. `IsPhysicalBRS` states measurability, finiteness of every counted intersection, and (1) for all real $x \leq y$, using the actual set cardinality and actual product Lebesgue volume. The covolume expression is the determinant calculated in Lemma 1; its interpretation as lattice covolume is a human verification, not an assumed theorem hidden in the definition.

The formal bridge includes:

- `canonical_internal`, `canonical_phys`, and `internal_unit_unique`: equations (4)–(5) and uniqueness.
- `canonical_cell`, `physical_block_reindex`: cell membership and the exact reindexing of accepted lattice parameters, for all nonnegative real α, β , all integers K , and all natural N . The general cell inequality follows from $1 + \alpha^2 + \beta^2 > \alpha + \beta$.
- `modelSet_block_eq`: the actual intersection of the model set with the aligned physical interval equals the explicitly finite set `blockPoints`. `canonical_phys_injective` and `blockPoints_card` verify that no duplicates change its cardinality.

- `physicalCount_eq_blockCount` and `blockCount_reduction`: actual model-set counting becomes the circle Birkhoff sum in (8).
- `modelSet_block_discrepancy`: the bound q for actual model-set counts, with area divided by covolume as the expected density.
- `literal_window_irrational_area` and `literal_window_areas_dense`: irrational product Lebesgue area, exclusion from half-integers, and density of the same rectangle family. The concrete declarations `literalGamma`, `literalWindow`, `literal_block_counterexamples`, and `literal_areas_dense` specialize to $\sqrt{2}, \sqrt{3}$.

The all-interval formal results, beyond the aligned-block bridge, are:

- `literal_window_discrepancy`: explicit finiteness and bound $q + 2$ on every real half-open physical interval, by inner/outer aligned-block inclusion and `Set.ncard_le_ncard`.
- `literal_window_isPhysicalBRS`: membership of each positive W_q in the unchanged physical counting predicate.
- `literal_volume_clause_false`: the headline negation

```
not (forall W : Set (Real * Real),
  IsPhysicalBRS (sqrt 2) (sqrt 3) W ->
    exists k : Int, (volume W).toReal = k / 2)
```

- `literal_physical_brs_areas_dense`: every open subinterval of $(0, 1)$ contains the area of a positive W_q , together with its physical BRS membership. `literal_area_spectrum_not_discrete` formally proves that the full physical BRS area spectrum is not a discrete subset of $\mathbb{R}$.

The only human bridge required for the literal-dimension disproof is Lemma 1: the displayed image of $\mathbb{Z}^3$ is a discrete full-rank lattice of covolume 6, its physical projection restricted to the lattice is injective, and its internal projection is dense in $\mathbb{R}^2$. No Lean theorem establishes these admissibility properties, and no additional axiom assumes them. The all-real-interval counting argument, BRS membership, universal half-integer negation, density in the physical BRS class, and topological nondiscreteness are Lean theorems. The optional closed-window and rational-polytope observations remain human arguments and are not needed for that negation.

7 Secondary internal-line reading

If one instead chooses a physical line and an internal line (total lattice dimension two), the circle results remain a separate counterexample. For every irrational rotation parameter r , $[0, \{qr\})$ is a circle BRS with irrational length for $q \geq 1$, and these lengths are dense in $(0, 1)$. The source retains `IsBRS`, `interval_counterexamples`, `circle_volume_clause_false`, and `interval_volumes_dense` for this reading.

For a concrete internal-line scheme, put $a = \{\sqrt{2}\} \in (0, 1)$ and

$$\Gamma_a = \{(k - am, m - ak) : k, m \in \mathbb{Z}\} \subset \mathbb{R} \times \mathbb{R}.$$

Its determinant is $c_a = 1 - a^2 > 0$, its physical projection is injective by irrationality, and its internal image $\mathbb{Z} + a\mathbb{Z}$ is dense. For $W = [0, a)$ there is a selected point precisely when $m = \lceil ak \rceil$ and $\{-ak\} < a$. Its physical coordinate is $t_k = c_a k - a \{-ak\}$, a displacement of less than a from

$c_a k$. The circle sum with rotation $-a$ and interval length $a = \{-(-a)\}$ has discrepancy at most one: its centered indicator is $\{x - a\} - \{x\}$, a coboundary for rotation $-a$ with transfer function $-\{x\}$. Comparing physical intervals with the grid changes only a bounded number of indices within distance a/c_a of their endpoints. Thus this secondary model also has bounded physical discrepancy and irrational internal length. This paragraph is a human proof.

The source also retains the supporting two-torus predicate `IsBRS2` and `rectangle_isBRS` for translation by (r, s) . That predicate is a Birkhoff-sum definition, and is not itself identified with a cut-and-project scheme. The rank-three counting proof in Sections 2–4 supplies the required primary bridge.

8 Verification, reproduction, and provenance

Lean and Mathlib are pinned to `v4.33.0`. The successful compiler output and axiom audits are recorded in `verification.txt`. The audited results depend only on

```
[propext, Classical.choice, Quot.sound]
```

There are no additional axioms, admissions, or native-decision shortcuts. The Lean source was checked against the pinned Mathlib environment.

```
cd lean4
lake exe cache get && lake build
cd ..
python3 verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The Python standard-library program checks exact rational examples of the coboundary, half-open endpoint conventions, physical-cell placement, model counting, and the arbitrary-interval estimate. These finite checks are supplemental and establish neither irrationality nor infinite density.

The transfer-function definition and four coboundary lemmas were adapted from `D5/S1/Phase/HeckeOstrowskiCoboundary.lean` in `trureturing`, immutable commit `1be30b25c4fe6cd46397090ca09b5696eb62ea78`. The Apache-2.0 license is included as `LICENSE.trureturing`. The namespace and imports were adapted; no `trureturing` import is needed.

References

- [1] H. Kesten, *On a conjecture of Erdős and Szűsz related to uniform distribution mod 1*, Acta Arithmetica **12** (1966), 193–212. <https://doi.org/10.4064/aa-12-2-193-212>.
- [2] S. Grepstad and N. Lev, *Sets of bounded discrepancy for multi-dimensional irrational rotation*, Geometric and Functional Analysis **25** (2015), 87–133. <https://doi.org/10.1007/s00039-015-0313-z>.
- [3] A. Haynes, M. Kelly, and H. Koivusalo, *Constructing bounded remainder sets and cut-and-project sets which are bounded distance to lattices*, Israel Journal of Mathematics **212** (2016), 189–201. <https://doi.org/10.1007/s11856-016-1283-z>.

Submitted by AlyciaBHZ on behalf of the Omega Institute (`trureturing` project, <https://github.com/the-omega-institute/trureturing>).